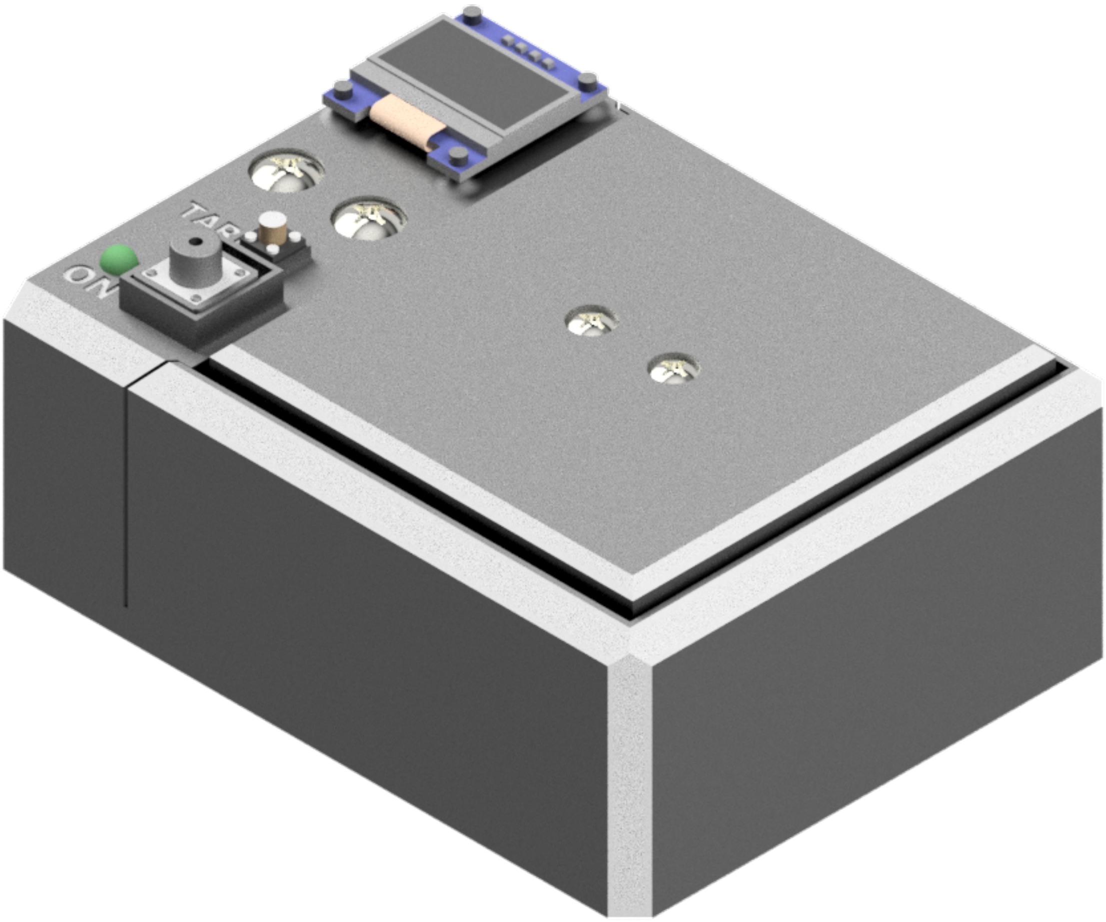
Scale Project

Final Project Documentation



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This document summarizes the project and describes the assumptions, design and operation of the device.

Contents

1 Introduction

1.1 Objective

The objective of the project was to design and build a simple scale for everyday use.

1.2 Operating Principle

The measurement principle is based on the linear relationship between stress and strain in the material from which the load cell is made. According to Hooke's law, the deformation of the beam is proportional to the applied force. Strain gauges attached to the beam respond to deformation by changing their resistance. Connected in a Wheatstone bridge, they convert this change into a voltage signal.

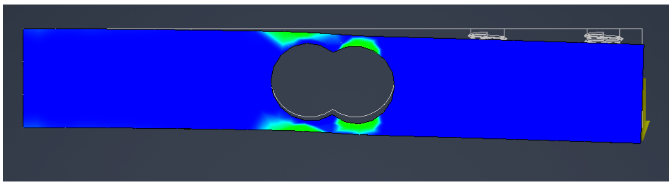


Figure 1.1: Visualization of load-cell deflection.

Because of the expected loads and material properties, load-cell deflection is very small and reaches only several tens of micrometres. The Wheatstone bridge output therefore requires amplification and conversion before it can be read by the microcontroller.

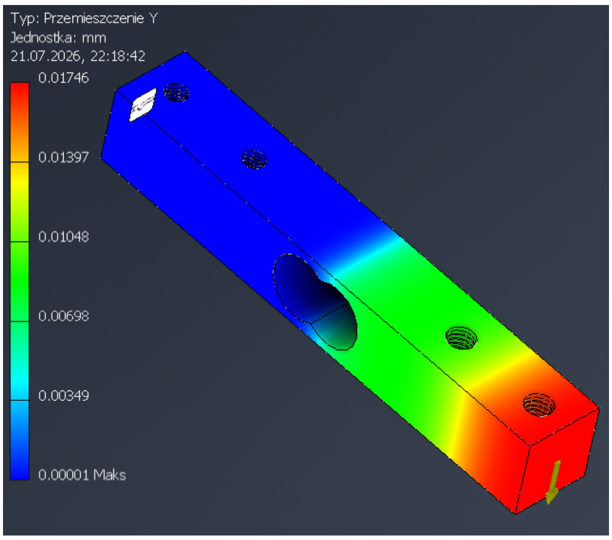


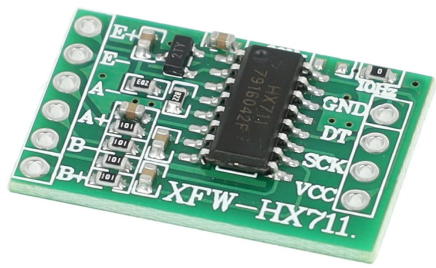
Figure 1.2: Example load-cell displacement.

2 Electronics

2.1 Selected Modules

After analyzing the project requirements, ready-made electronic modules were selected.

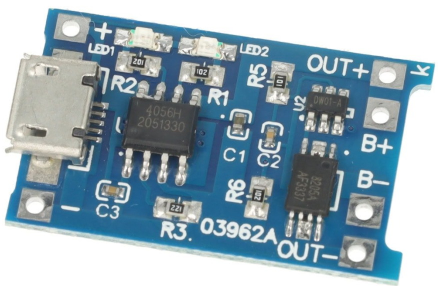
(a) Microcontroller.



(b) Load-cell amplifier.

The ATtiny85 microcontroller was selected initially. However, because the display had to be supported, it was replaced with the more capable DFRduino Nano development board [1], equipped with an ATmega328 microcontroller. Its availability, sufficient capabilities and favorable purchase price of PLN 20 were the main reasons for this choice.

The HX711 [2], a 24-bit analog-to-digital converter with an integrated programmable gain amplifier, was used to process the Wheatstone bridge signal.

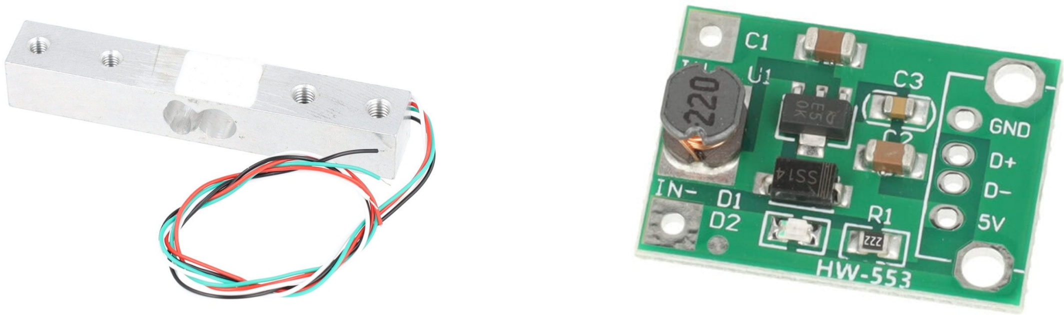


(a) Cell charging and protection module.

(b) OLED display.

Due to the portable nature of the device, battery power was selected. A module based on the TP4056 [3] is responsible for charging and protecting the cell.

The results are shown on a 0.96-inch OLED display [4], which communicates with the microcontroller through the I2C bus. The modules require a 5 V supply, while the cell voltage is between 3.3 and 4.2 V. A step-up converter was therefore used to provide 5 V [5].



(a) Load cell.

(b) Step-up converter.

The main measuring component is a load cell [6] with a range of up to 10 kg. Its operating principle is explained in Section 1.2. The range was selected for the expected loads. In a typical kitchen scale, a load cell with a range of 1–2 kg would provide better measurement resolution.

2.2 Main Board

The modules were soldered to a prototype board. ARK screw terminals were also installed to simplify wire connections.

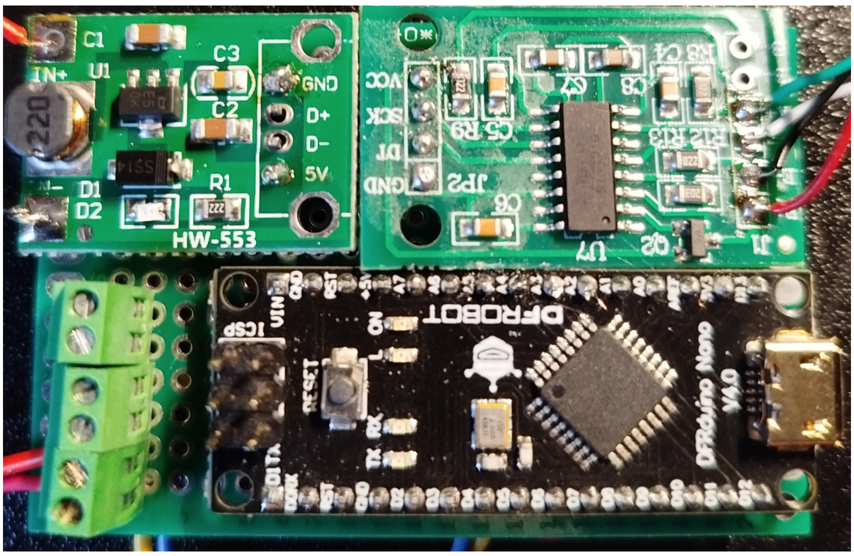


Figure 2.4: Assembled main board.

3 Mechanical Design

The enclosure was designed as a three-part bolted structure consisting of a base, an interface bridge and a weighing platform. It arranges the components compactly and safely while allowing the load cell to operate freely. Later iterations improved the mounting of the threaded inserts and reduced the platform outline to eliminate contact with the base walls. The display and LED were press-fitted into the enclosure after local heating of the plastic.

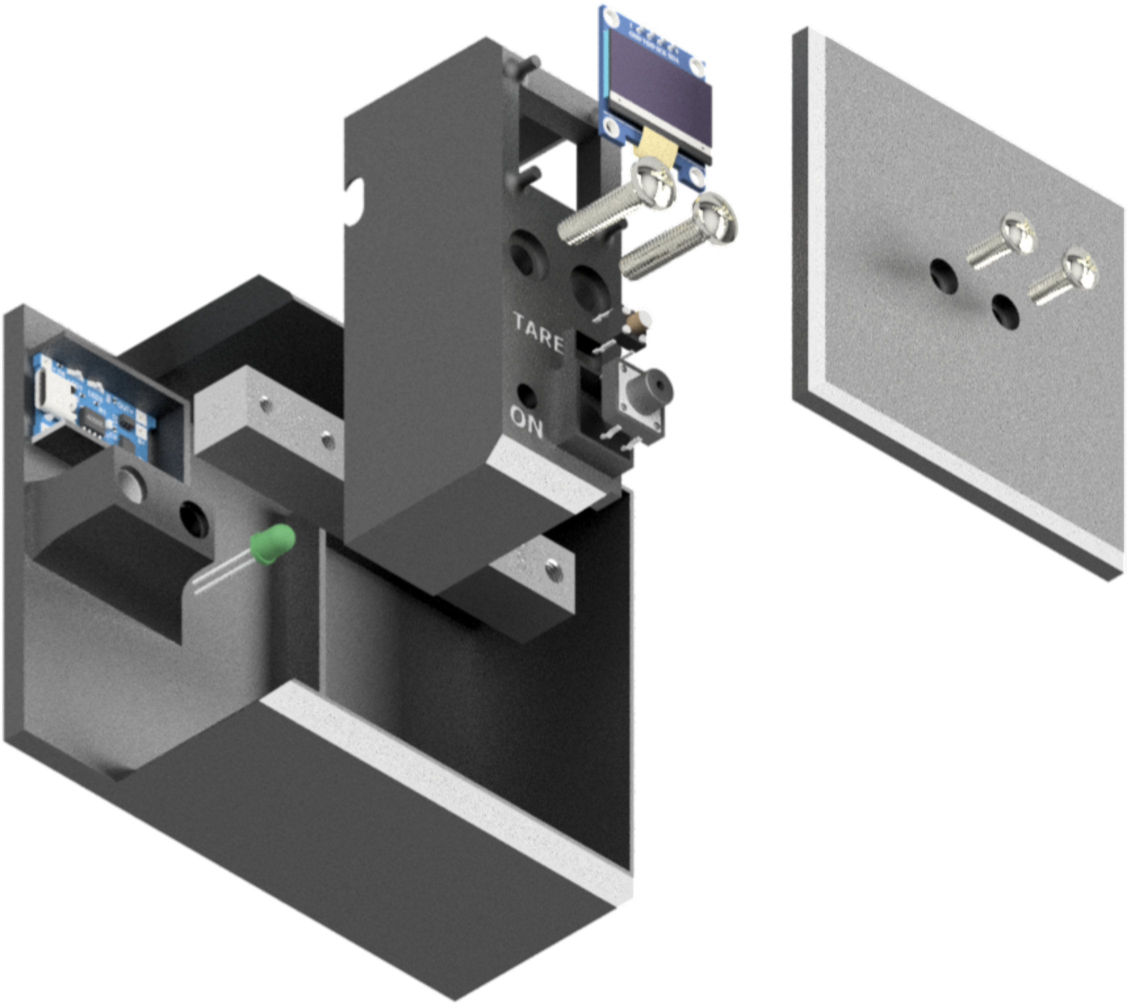


Figure 3.1: Device assembly.

The base contains the main board, TP4056 module, cell and load cell, which is mounted using two M5 heat-set threaded inserts. A thicker floor and rounded column edges increase the stiffness of the structure.

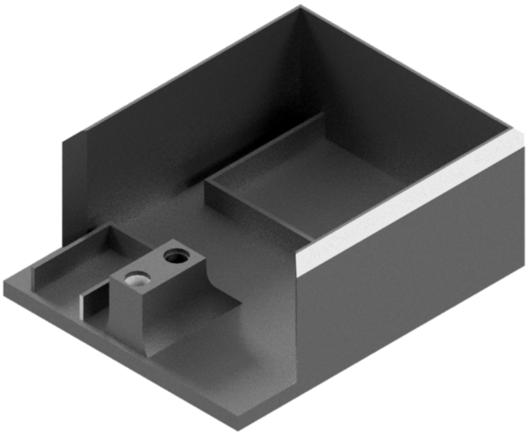


Figure 3.2: Device base.

The interface bridge contains the display, LED, tare button and power button. It includes holes for M5 screws, four display mounting posts and a cut-out providing access to the USB port of the TP4056 module. Its chamfered walls were matched to the base.

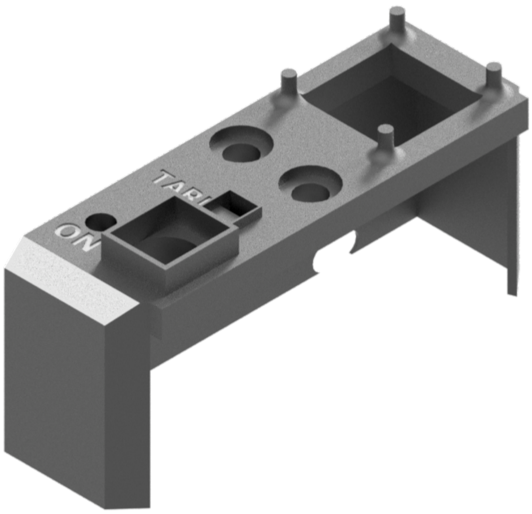


Figure 3.3: Interface bridge.

The weighing platform has a working area of $73 \times 73\,\mathrm{mm}$ and chamfered upper corners. It is attached to the load cell with two M4 screws. Its reduced outline prevents contact with the enclosure when the beam deflects.

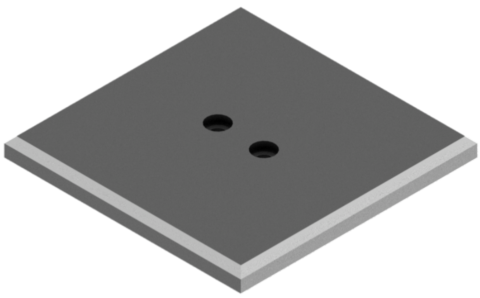


Figure 3.4: Weighing platform.

4 Software

4.1 Preliminary Calibration

A calibration factor passed as an argument to the `set_scale()` function was used to convert the raw HX711 signal into mass. Its initial value was determined using everyday objects and a hammer with a known mass of 590 g. The calibration factor was calculated from:

$$Scale_0 = \frac{N_{\text{obc}} - N_0}{m_{\text{ref}}}, \tag{4.1}$$

where N_{obc} is the converter reading after loading the cell, N_0 is the unloaded scale reading and m_{ref} is the mass of the reference object. For the values obtained experimentally, the factor was:

$$Scale_0 = \frac{181224 - 44012}{590} \approx 232.56. \tag{4.2}$$

During the first measurements, this value was passed to the `scale.set_scale()` function. The HX711 library uses it to convert the difference between the current reading and the value determined during taring into a result expressed in grams.

Taring was performed automatically after the device was started and after the button was pressed. The `tare()` function determined the zero level from 20 samples. At the same time, the value stored by the software filter was reset.

To reduce short-term fluctuations, the `getWeightResult()` function calculated the average of five converter readings and then applied an exponential filter:

$$m_f(k) = \alpha m(k) + (1 - \alpha) m_f(k-1), \tag{4.3}$$

where $m(k)$ is the current measurement result, $m_f(k)$ is the filtered result and the α factor was set to 0.8. The result was shown on the OLED display. Values below 1 g were set to zero in software, while results above 1000 g were displayed in kilograms.

The preliminary validation results described in Section 5.1 showed systematic under-reading of the measured mass. Corrected factor values were calculated for each load, and their average was approximately 229.99. The following rounded value was therefore used in the program:

```
static float scaleFactor = 230.0f;
```

Reducing the factor from 232.56 to 230.00 increased the calculated result by approximately 1.11%, compensating for the under-reading observed during preliminary validation. This factor was used until a more accurate calibration with a set of test weights was performed.

4.2 Final Calibration

The $Scale = 230.00$ factor was determined by comparison with a commercial kitchen scale. Because its accuracy and calibration status were unknown, this value was treated as the result of preliminary calibration. To determine the sensitivity of the device more accurately, another series of tests was performed using test weights with masses from 10 to 500 g.

Before the measurements, the device was left switched on for approximately 10 minutes to reduce the influence of temperature changes in the load cell and HX711 amplifier. Before each measurement point, the load was removed from the platform and the scale was tared.

For each weight, the difference between the raw converter reading under load and the level determined during taring was calculated:

$$x_i = N_{\mathrm{obc},i} - N_{0,i}.$$

A linear conversion model was assumed:

$$x_i \approx S_k m_{\mathrm{ref},i},$$

where $m_{\mathrm{ref},i}$ is the declared mass of the weight and S_k is the final calibration factor to be determined. Its value was calculated by the least-squares method, assuming a straight line passing through the origin:

$$S_k = \frac{\sum_{i=1}^{n} m_{\mathrm{ref},i} x_i}{\sum_{i=1}^{n} m_{\mathrm{ref},i}^2}.$$

The measurements gave:

$$S_k = 232.3866.$$

This value was rounded to two decimal places and stored in the final version of the program:

```
static float scaleFactor = 232.39f;
```

The final calibration result differs from the value of 230.00 determined during preliminary validation. The preliminary factor was based on the readings of a commercial kitchen scale and was affected by zero-level drift. The use of test weights, device warm-up and taring before each point made it possible to determine the sensitivity of the system more accurately.

Increasing the factor from 230.00 to 232.39 proportionally reduces the result returned by the `get_units()` function. This prevents the over-reading that would occur if the factor of 230.00 were used under stabilized measurement conditions.

4.2.1 Linearity Assessment

Before the final calibration, possible nonlinearity of the load-cell characteristic was considered. It could result from beam deformation, the mounting method or the transfer of the load through the platform. For this reason, the use of a quadratic correction was analyzed:

$$m_{\mathrm{corr}} = K_1 m + K_2 m^2.$$

The initial measurement series could suggest slight curvature of the characteristic. However, most of this difference disappeared when the scale was tared before each measurement point. This indicates that its main sources were zero-level drift and load-cell creep rather than actual nonlinearity of its sensitivity.

After applying a single linear factor of $S_k = 232.39$, the maximum absolute error for weights from 10 to 500 g was approximately 0.19 g, while the RMSE was approximately 0.116 g. Fitting a quadratic model reduced the RMSE to approximately 0.071 g, an improvement of only about 0.045 g. This improvement was small compared with the variation observed between measurement series.

In addition, the errors of the linear model changed sign and did not form a clear monotonic trend related to the applied load. Using two factors, K_1 and K_2, would therefore increase the risk of fitting the function to measurement noise, zero drift and errors in the placement of the weights. For this reason, quadratic correction was not used in the range covered by the calibration weights, i.e. from 10 to 500 g, and the linear model was retained.

This result confirms linearity only within the tested range. It does not prove that the load cell is linear over its entire nominal range of 10 kg.

4.2.2 Extended-Range Correction

Certified calibration weights were not available for loads above 500 g. An additional comparison with a commercial kitchen scale was therefore performed. As the mass increased, a systematic under-reading of the constructed scale was observed.

To partially compensate for this difference, a simple continuous piecewise-linear correction was applied:

$$m_{\mathrm{corr}} = \begin{cases} m, & m \le 500 \text{ g}, \\ m + c(m - 500), & m > 500 \text{ g}, \end{cases}$$

where m is the result obtained after applying the factor $S_k = 232.39$.

The c factor was determined by the least-squares method. The following variable was defined:

$$z_i = \max(m_i - 500, 0),$$

and then:

$$c = \frac{\sum_i z_i \left(m_{\text{kuch},i} - m_i\right)}{\sum_i z_i^2} \approx 0.003965.$$

The final function implemented in the program is:

```
float calibrationCorrection(float grams)
{
  if (grams <= 500.0f)
    return grams;

  return grams + 0.003965f * (grams - 500.0f);
}
```

The advantage of this function is that the result obtained using calibration weights up to 500 g is preserved and no discontinuity occurs at the transition point. This is neither a quadratic correction nor a complete metrological calibration of the extended range. It is only an empirical compensation of the difference observed relative to a kitchen scale whose accuracy and calibration status were unknown.

4.3 Filtering and Reading Stabilization

The rate at which new results can be obtained is limited mainly by the sampling rate of the HX711. The `delay(1)` instruction used in the main program loop is not the main factor determining the measurement time.

In the final version of the program, the `get_units(2)` function calculates the average of two consecutive samples. This produces a result faster than averaging five samples while still providing basic noise reduction.

An exponential filter is then applied. Its factor is selected according to the change in the reading:

$$m_f(k) = \alpha(k)m(k) + [1 - \alpha(k)]\, m_f(k-1),$$

gdzie:

$$\alpha(k) = \begin{cases} 0.85, & |m(k) - m_f(k-1)| > 1 \text{ g}, \\ 0.10, & |m(k) - m_f(k-1)| \le 1 \text{ g}. \end{cases}$$

A high value of $\alpha = 0.85$ means that 85% of the new result is included in the current iteration. The filter therefore responds quickly when an object is placed on or removed from the platform. Once the reading has stabilized, $\alpha = 0.10$ is used to provide stronger attenuation of small changes and converter noise.

During taring, the zero level is determined from 25 samples:

```
scale.tare(25);
```

```
filteredWeight = 0;
```

Resetting the filter variable prevents the previous reading from affecting the next measurement.

The result is displayed with two decimal places. However, the number of displayed digits defines only the presentation format and should not be interpreted as the measurement accuracy of the device.

5 Validation

5.1 Preliminary Validation

Preliminary validation was performed by comparing the readings of the constructed device with the results of a commercial kitchen scale. Because certified calibration weights were not available at this stage, the kitchen scale was used as the reference device. Therefore, the test did not constitute a complete metrological calibration.

Five objects with masses ranging from approximately 190 g to 1.84 kg were used. Each object was measured alternately using the constructed scale and the reference scale. Three repetitions were performed for each load, and the average readings were then calculated.

Table 5.1: Comparison of average measurement results.

No.	Average measured [g]	Average reference [g]	Difference [g]	Error [%]
1	188.23	191.00	2.77	1.45
2	1822.67	1838.33	15.67	0.85
3	1552.00	1566.00	14.00	0.89
4	883.73	893.67	9.93	1.11
5	513.60	520.00	6.40	1.23

For all tested objects, the reading of the constructed scale was lower than that of the reference device. The relative error ranged from 0.85% to 1.45%, with an average value of approximately 1.11%. The similar direction of the deviation across the tested range mainly indicates an inaccurate calibration factor rather than a random error.

For each load, a corrected factor can be calculated from:

$$Scale_i = Scale_0 \frac{\overline{m}_i}{\overline{m}_{\mathrm{ref},i}}. \tag{5.1}$$

For the initial factor $Scale_0 = 232.56$, the average factor obtained from the five series was:

$$\overline{Scale} = 229.99.$$

For this reason, the following rounded value was used in the program:

```
static float scaleFactor = 230.0f;
```

Reducing the factor from 232.56 to 230.00 increased the calculated result by approximately 1.11%, compensating for the systematic under-reading observed relative to the kitchen scale.

This result was treated as a preliminary software correction. Because a certified reference device was not available, the factor had to be determined again using a set of test weights. These tests are described in the final calibration section.

5.2 Final Validation Using Calibration Weights

The final measurements were performed at ten points covering nominal masses from 10 to 500 g. The device was left switched on for approximately 10 minutes before the measurements. Before each point, the load was removed, the scale was tared and the weights were then placed in the center of the platform.

Table 5.2 presents the values obtained after applying the final factor $S_k = 232.39$. The error was defined as:

$$e_i = m_{k,i} - m_{\mathrm{ref},i}.$$

Table 5.2: Assessment of the final calibration-factor fit.

Nominal mass [g]	**Value for $S_k = 232.39$ [g]**	**Error [g]**
10	10.02	0.02
20	19.96	-0.04
40	40.06	0.06
60	60.09	0.09
80	79.81	-0.19
100	99.91	-0.09
150	149.87	-0.13
200	199.83	-0.17
400	399.96	-0.04
500	500.17	0.17

The mean error, mean absolute error and root mean square error were used for a quantitative assessment of the fit:

$$\overline{e} = \frac{1}{n}\sum_{i=1}^{n} e_i,$$

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |e_i|,$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} e_i^2}.$$

For the data in the table:

$$\overline{e} = -0.031\ \mathrm{g}, \qquad \mathrm{MAE} = 0.099\ \mathrm{g}, \qquad \mathrm{RMSE} = 0.116\ \mathrm{g}.$$

The maximum absolute error was approximately:

$$|e|_{\max} = 0.189 \text{ g}.$$

The small mean error indicates that there is no clear systematic over-reading or under-reading after the factor is changed. The changing signs of consecutive errors also do not indicate clear nonlinearity within the tested range.

The presented values were calculated from the same data series that was used to determine the S_k factor. They therefore assess the quality of the fit rather than provide fully independent validation. Independent confirmation of accuracy would require an additional measurement series after storing the factor of 232.39 in the software.

5.3 Zero-Drift Assessment

In longer series where taring was performed only before the start of the test, a gradual shift in the reading was observed. This effect could appear similar to nonlinearity or hysteresis, particularly when comparing series performed with increasing and decreasing mass.

After taring before each measurement point, the shift was greatly reduced and the readings again approached the reference values. This indicates that the main sources of the changes were zero-level drift, temperature dependence and load-cell creep.

Taring compensates for the additive component of the error present at the time it is performed, but it does not change the sensitivity of the system or replace a correct calibration factor. It also does not guarantee the removal of changes that occur after the load is applied. The following measurement procedure was therefore adopted:

1. leave the device switched on for approximately 10 minutes,
2. make sure that the platform is unloaded,
3. tare the scale immediately before the measurement,
4. place the object as close to the center of the platform as possible,
5. wait for the reading to stabilize.

5.4 Extended-Range Comparison

An additional assessment for masses above 500 g was performed by comparison with a commercial kitchen scale. The results previously obtained using the constructed scale were recalculated using the final factor $S_k = 232.39$ and the piecewise-linear correction.

Relative to the kitchen-scale readings:

$$\text{MAE} = 1.58 \text{ g}, \qquad \text{RMSE} = 1.89 \text{ g},$$

Table 5.3: Extended-range comparison after application of the final correction.

Object	Kitchen scale [g]	Corrected value [g]	Difference [g]
1	340	340.57	0.57
2	1290	1287.15	-2.85
3	1828	1824.96	-3.04
4	1554	1551.02	-2.98
5	771	770.59	-0.41
6	2425	2425.21	0.21
7	594	592.73	-1.27
8	3810	3812.03	2.03
9	3038	3039.56	1.56
10	2129	2128.11	-0.89

while the maximum absolute difference was approximately 3.04 g.

The results indicate that the correction reduced the systematic under-reading observed for larger loads. However, this comparison should be treated only as a verification test. The kitchen scale was not a certified reference device and showed values approximately 1 g higher when the 150, 200 and 400 g weights were measured. In addition, the same results were used both to determine the correction factor and to calculate the presented errors.

Complete validation of the extended range would require certified standards with masses above 500 g and independent measurement series.

5.5 Validation Conclusions

Preliminary validation against a commercial kitchen scale led to the selection of the factor $Scale = 230.00$. This value reduced the initially observed under-reading, but it was only an approximate result because of the accuracy limitations of the reference device.

Final calibration with a set of test weights, after warming up the device and taring before each point, made it possible to determine a more accurate factor:

$$\boxed{S_k = 232.39}.$$

Within the range from 10 to 500 g, the device characteristic can be considered linear with accuracy sufficient for the intended application. A quadratic correction was analyzed, but its effect on the error was small and did not justify increasing the complexity of the model.

For masses above 500 g, an additional piecewise-linear correction was applied. Its effect was assessed by comparison with a kitchen scale, but because no certified standard was available, this result cannot be treated as a complete metrological calibration.

The combination of two-sample averaging, a dynamic exponential filter and taring based on 25 samples provided a compromise between response time and reading stability. Taring before the measurement effectively reduced the observed zero drift, but it did not replace calibration of the device sensitivity.

Bibliography

[1] Botland. *DFRduino Nano V3.1 – kompatybilny z Arduino Nano*. Botland. URL: https://botland.com.pl/plytki-zgodne-z-arduino-dfrobot/3035-dfrduino-nano-v40-kompatybilny-z-arduino-nano-6959420900398.html (visited on 07/19/2026).

[2] Sklep msalamon. *Wzmacniacz belki tensometrycznej HX711*. Sklep msalamon. URL: https://sklep.msalamon.pl/produkt/wzmacniacz-belki-tensometrycznej-hx711/ (visited on 07/19/2026).

[3] Botland. *Ładowarka Li-Ion HW-373 V1.2.1 TP4056 – pojedyncza cela 1S 3,7 V, USB-C, z zabezpieczeniami*. Botland. URL: https://botland.com.pl/ladowarki-elektryczne/16979-ladowarka-li-ion-hw-373-v121-tp4056-pojedyncza-cela-1s-37v-usb-typ-c-z-zabezpieczeniami-5904422326708.html (visited on 07/19/2026).

[4] Botland. *Wyświetlacz graficzny OLED 1,3 cala, 128x64 px, I2C V2 – niebieskie znaki*. Botland. URL: https://botland.com.pl/wyswietlacze-oled/8246-wyswietlacz-oled-niebieski-graficzny-13-128x64px-i2c-v2-niebieskie-znaki-5904422311339.html (visited on 07/19/2026).

[5] Sklep msalamon. *Mini przetwornica step-up 1–5 V do 5 V 600 mA*. Sklep msalamon. URL: https://sklep.msalamon.pl/produkt/mini-przetwornica-step-up-1-5v-do-5v-600-ma/ (visited on 07/22/2026).

[6] Sklep msalamon. *Belka tensometryczna do 10 kg / 100 N*. Sklep msalamon. URL: https://sklep.msalamon.pl/produkt/belka-tensometryczna-do-10-kg-100-n/ (visited on 07/19/2026).